Siddhartha Prasad

siddhartha.a.prasad@gmail.com | siddharthaprasad.com

Bringing the machinery of programming languages and formal methods to bear on the cognitive work of programming.

Education

Ph.D. in Computer Science, **Brown University**, Providence, RI *Jan 2022 – May 2027 (expected)*
Advisor: Shriram Krishnamurthi

B.S. in Computer Science and Mathematics, **Tufts University**, Medford, MA *May 2016*
Advisor: Kathleen Fisher · *Summa cum Laude; High Thesis Honors*

Employment

Apple · Research Intern · Human-Centered Machine Learning · Seattle, WA *May – Sep 2023*

- Designed a formal-methods-inspired technique that prompts an LLM to evaluate candidate outputs against task-specific quality criteria and iteratively refine them without fine-tuning.
- Applied the technique in ContextQ, a system that generates questions to support parent-child dialogue during shared reading.

Microsoft Redmond, WA

Software Engineer II · Applied AI *Apr 2018 – Sep 2021*

- Built the container platform for Azure Cognitive Services, enabling customers with sensitive data to run AI workloads on-premises, at the edge, or in fully disconnected environments.
- Designed a deployment framework that allowed the same Docker containers to run across Azure-hosted and customer-managed environments, abstracting differences in CPU architecture, connectivity, API-key validation, logging, and storage.
- Operated production infrastructure for the cloud service, including on-call responsibility.

Software Engineer · Developer Ecosystem Platform *Aug 2016 – Mar 2018*

- Designed and shipped XAML input APIs across Windows 10, Windows Phone, and Xbox, giving developers a consistent model for keyboard, gamepad, and ink interaction.
- Built accessibility and navigation features, including XAML access keys, so applications could expose commands and behave consistently across keyboard and gamepad input.

Software Engineering Intern · Developer Ecosystem Platform *May – Aug 2015*

INRIA · Research Intern · Parsifal · Saclay, France *Jun – Jul 2014*

- Researched correctness certificates for first-order term-rewriting systems.

Publications

Diagramming Program Values by Spatial Refinement
Siddhartha Prasad, Michael Tu, Karan Kashyap, Tim Nelson, Shriram Krishnamurthi
ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI), 2026

Meaningful Human-in-the-Loop Checking of GenAI Synthesis for Restricted Languages
Siddhartha Prasad, Skyler Austen, Kathi Fisler, Shriram Krishnamurthi
European Conference on Object-Oriented Programming (ECOOP), 2026

hS: Speculative Script Reordering at Subprocess Granularity
Georgios Liargkovas, Di Jin, Tianyu (Ezri) Zhu, Dan Liu, A. Bolun Thompson, Anirudh Narsipur, Seong-Heon Jung, *Siddhartha Prasad*, Diomidis Spinellis, Michael Greenberg, Konstantinos Kallas, Nikos Vasilakis
USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2026

A Misconception-Driven Adaptive Tutor for Linear Temporal Logic
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi
International Conference on Computer Aided Verification (CAV), 2025
Distinguished Paper Award

Lightweight Diagramming for Lightweight Formal Methods: A Grounded Language Design
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi
European Conference on Object-Oriented Programming (ECOOP), 2025
Distinguished Paper Award; Distinguished Artifact Award

Misconceptions in Finite-Trace and Infinite-Trace Linear Temporal Logic
Ben Greenman, *Siddhartha Prasad*, Antonio Di Stasio, Shufang Zhu, Giuseppe De Giacomo, Shriram Krishnamurthi, Marco Montali, Tim Nelson, Milda Zizyte
International Symposium on Formal Methods (FM), 2024

ContextQ: Generated Questions to Support Meaningful Parent-Child Dialogue While Co-Reading
Griffin Dietz Smith, *Siddhartha Prasad*, Matt J Davidson, Leah Findlater, R Benjamin Shapiro
ACM Interaction Design and Children Conference (IDC), 2024

Forge: A Tool and Language for Teaching Formal Methods
Tim Nelson, Ben Greenman, *Siddhartha Prasad*, Tristan Dyer, Ethan Bove, Qianfan Chen, Charles Cutting, Thomas Del Vecchio, Sidney LeVine, Julianne Rudner, Ben Ryjikov, Alexander Varga, Andrew Wagner, Luke West, Shriram Krishnamurthi
ACM SIGPLAN Conference on Object-Oriented Programming Systems, Languages & Applications (OOPSLA), 2024

Conceptual Mutation Testing for Student Programming Misconceptions
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi
The Art, Science, and Engineering of Programming, 2024

Generating Programs Trivially: Student Use of Large Language Models
Siddhartha Prasad, Ben Greenman, Tim Nelson, Shriram Krishnamurthi
ACM Conference on Global Computing Education (CompEd), 2023

Making Hay from Wheats: A Classsourcing Method to Identify Misconceptions
Siddhartha Prasad, Ben Greenman, Tim Nelson, John Wrenn, Shriram Krishnamurthi
Koli Calling International Conference on Computing Education Research, 2022

Large-Scale Intelligent Microservices
Mark Hamilton, Nick Gonsalves, Christina Lee, Anand Raman, Brendan Walsh, *Siddhartha Prasad*, Dalitso Banda, Lucy Zhang, Lei Zhang, William T Freeman
IEEE International Conference on Big Data, 2020

Invited Talks

Eventually, Understanding: Operationalizing Misconceptions in LTL
Brandeis University — Apr 2026
Indian Association for Research in Computing Science (IARCS) Verification Seminar Series — Mar 2026

Lightweight Diagramming by Spatial Specification
MIT HCI Seminar — Jan 2026

Understanding Problems Before Programming
IIT Gandhinagar — Jan 2024
Ashoka University — Mar 2023

Teaching

Teaching Assistant, **Brown University**
Logic for Systems — 2022–2026
Programming Languages — 2025

Teaching Assistant, **Tufts University**
Programming Languages — 2015–2016
Algorithms — 2014

Mentorship

Summer 2026	**Owen Duckham**, Diagramming for Lean Brown University
Summer 2025	**Michael Tu**, Spatial refinement for diagramming program values Brown University
Winter 2019	**Patrick Kao**, Windows emulation on Linux Microsoft

Service & Volunteering

2026	Artifact Reviewer, The Art, Science, and Engineering of Programming
2017	Volunteer Alumni Interviewer, Tufts Admissions Network

Affiliations & Certifications

Sheridan Teaching Seminar Certificate · Brown University
Tau Beta Pi, The Engineering Honor Society · Tufts University